

1. Questions

The ratio of the present ages of A to that of B is 9:14, and the ratio of the present ages of B to that of C is 4:5, respectively. If the difference between the present ages of A and C is neither less than 15 years nor more than 35 years but a multiple of 17, What is the possible average age of A and B?

I: 23 years

II: 69 years

III: 46 years

- a. Only I
- b. Only II
- c. Both I and III
- d. Both II and III
- e. Only III

2. Questions

Arun lends Rs. 21000 to John at $R\%$ SI per month. After 2 years, Arun received a total amount of Rs. 28560. If Arun invested the whole amount received from John in a scheme namely A at $10R\%$ compounded annually for 2 years, Find the difference between the interest received from scheme A and the interest received from John.

- a. Rs. 1650.6
- b. Rs. 1666.6
- c. Rs. 1460.6
- d. Rs. 1550.6
- e. Rs. 1160.6

3. Questions

The salaries of Raja and Ravi in 2018 was 5:4, respectively. The ratio of the salary of Raja in 2019 to that in 2018 is 4:3, and the ratio of the salary of Ravi in 2019 to that in 2018 is 7:6, respectively. If the sum of $\frac{1}{5}$ of the Raja's salary and $\frac{1}{7}$ of the Ravi's salary in 2019 is Rs. 6,000. Which of the following statements is true according to the difference between the salaries of Raja and Ravi in 2018?

- I). Multiple of 8
 - II). Multiple of 1000
 - III). Multiple of 2000
 - IV). Multiple of 3
- a. III and IV

- b. I, II, and IV
- c. I, II, and III
- d. All of them
- e. II and IV

4. Questions

A, B, C, and D to complete a work in factors of 8 ($A > B > C > D$) in days. If D only started the work at half of his efficiency, after 1 day, D left the work and A, B, and C (C works with 25% of his efficiency while A and B work with 100% of their efficiency) started working together to complete the remaining work. Find the total time taken to complete the work.

- a. 1 day
- b. 4 days
- c. 3 days
- d. 2 days
- e. 6 days

5. Questions

The time taken by a man to run at 2.5 m/sec in ____ km is ____ minutes. If the man increased the speed by 1 km/hr, then the time taken to cover 2 km in ____ seconds. Which of the following statements is true?

- I). 1.35, 9, and 720
 - II). 2.25, 15, and 720
 - III). 1.125, 9, and 360
- a. Both I and II
 - b. Only II
 - c. Both II and III
 - d. Only I
 - e. All of them

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6. Questions

Study the following information carefully and answer the questions given below:

The given table shows the total number of earbuds headphones available, the percentage of the number of over-ear headphones available out of the total number of earbuds headphones available, and the ratio of the number of over-ear headphones available to the number of in-ear headphones available in four different shops namely A, B, C and D.

Shop	The number of earbuds headphones available	The percentage of the number of over-ear headphones available out of the total number of earbuds headphones available	The ratio of the number of over-ear headphones available to the number of in-ear headphones available
A	$X + 40$	60%	12: 7
B	$Y - 45$	(40% of X)%	17: 10
C	$X - 20$	55%	11: 6
D	Y	49%	7: 5

Notes:

i). The total number of headphones available = The number of earbuds headphones available + The number of over-ear headphones available + The number of in-ear headphones available

ii). The number of in-ear headphones available in shop A is 21 less than that of shop D.

iii). The total number of headphones available in shop C is 333.

If the difference between the number of in-ear headphones available in shop A and the number of over-ear headphones available in shop B is 'P,' Which of the following statements is true?

- The number of earbuds headphones available in shop A is $2P$
- The number of earbuds headphones available in shop C is $1.5P$
- The number of in-ear headphones available in shop B is P
- Both (a) and (b)
- All (a), (b), and (c)

7. Questions

If 60%, 75%, and 80% of the number of earbuds headphones, over-ear headphones, and in-ear headphones respectively are sold in shop B, Find the total number of unsold headphones in shop B.

- 177
- 198
- 163
- 159
- None of the above

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8. Questions

The following questions have two quantities, Quantity I and Quantity II. You have to determine the relationship between them and give the answer as

Quantity I: The average of the number of in-ear headphones available in shops A, B, and C together.

Quantity II: If 83.33% of the total number of headphones in shop D are in good condition, the others are defective. Find the defective headphones in shop D.

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

9. Questions

If the ratio of earbuds headphones sold to over-ear headphones sold in shop D is 6:1 and the sum of over-ear and in-ear headphones sold in shop D is 119, If the rate per headphone for earbuds, over-ear, and in-ear is Rs. 500, Rs. 800, and Rs. 850, respectively and the total selling amount is Rs. 204400. The total number of headphones sold is approximately what percentage of the total headphones available in shop D.

- a. 50%
- b. 70%
- c. 80%
- d. 60%
- e. 40%

10. Questions

If the total number of headphones available in shop E is 33.33% more than that of Shop C, the number of earbuds headphones available in shop E is $(0.5X - 20)\%$ of that in Shop A, and the number of in-ear headphones available in shop E is 22.22% more than that of shop C, find the number of over-ear headphones available in shop E.

- a. $Y - 26$
- b. $X + 26$
- c. $0.5Y + 12$
- d. $X + 12$
- e. $X - 14$

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11. Questions

Study the following information carefully and answer the questions given below:

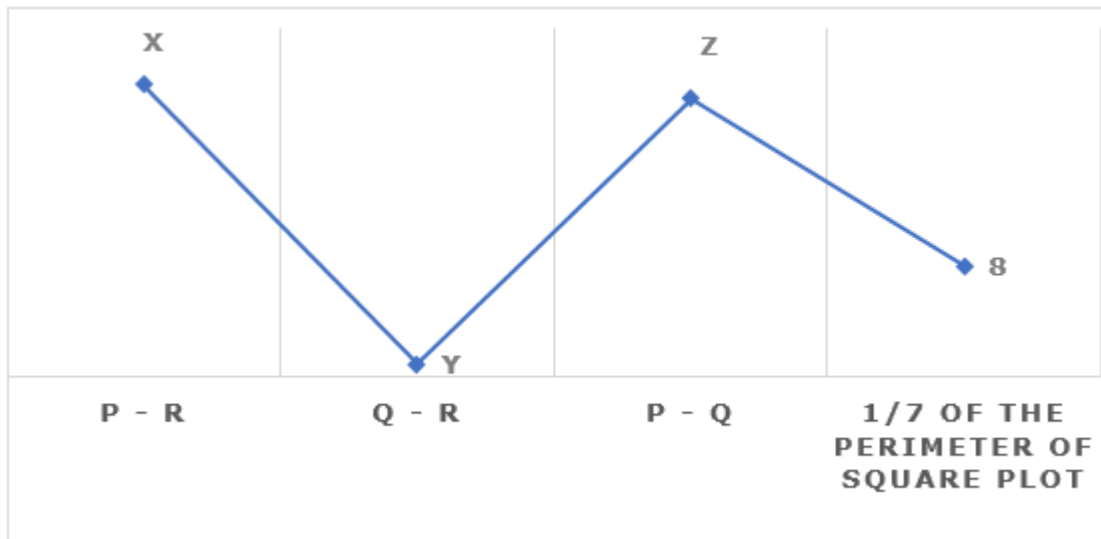
There is a square plot, and this plot is divided into two equal parts, A and B. The part B contains an equatorial triangle-shaped garden and the circle-shaped walking spot, and part A is full of residential purposes.

The given line graph shows the difference between the perimeter of the equatorial triangle-shaped garden and that of part A, the difference between the perimeter of the equatorial triangle-shaped garden and that of the perimeter of the circle-shaped walking spot, the difference between the perimeter of the circle-shaped walking spot and that of part A, and 1/7th of the perimeter of the square plot.

P ---> the perimeter of part A

Q ---> the perimeter of the circle-shaped walking spot

R ---> the perimeter of the equatorial triangle-shaped garden



Note:

- i). The base of the equatorial triangle-shaped garden is half of the side of the square plot.
- ii). All measurements in meters.
- iii). The breadth of part B is the same as the diameter of the circle-shaped walking spot.

Find the remaining area of part B (except the circle-shaped walking spot and the equatorial triangle-shaped garden).

Note: The value of $\sqrt{3}$ is 1.7

- a. 38.675 m²
- b. 42.65 m²
- c. 28.275 m²
- d. 31.25 m²
- e. 48.75 m²

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12. Questions

If a cylinder-shaped water tank is fixed in nearby the square plot. The radius of the water tank is 50% more than the length of part A, and the volume of the tank is 33264 m³. Find the height of the

water tank.

- a. 21 m
- b. 22 m
- c. 28 m
- d. 32 m
- e. 24 m

13. Questions

If the residential area of part A is divided into 4 housing plots. The breadth of part A is equally divided, and the ratio of the length of part A is divided as larger and smaller, which is 4:3 respectively. Find the area of one of the larger housing plots in part A.

- a. 21 m^2
- b. 42 m^2
- c. 24 m^2
- d. 28 m^2
- e. 56 m^2

14. Questions

What is the ratio of the area of part A to the area of the circle-shaped walking spot in part B in the square plot?

- a. 14: 11
- b. 28: 11
- c. 23: 12
- d. 11: 4
- e. 14: 3

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15. Questions

If the side of the square plot is increased by 25%, then find what percentage of the area increased in the square plot.

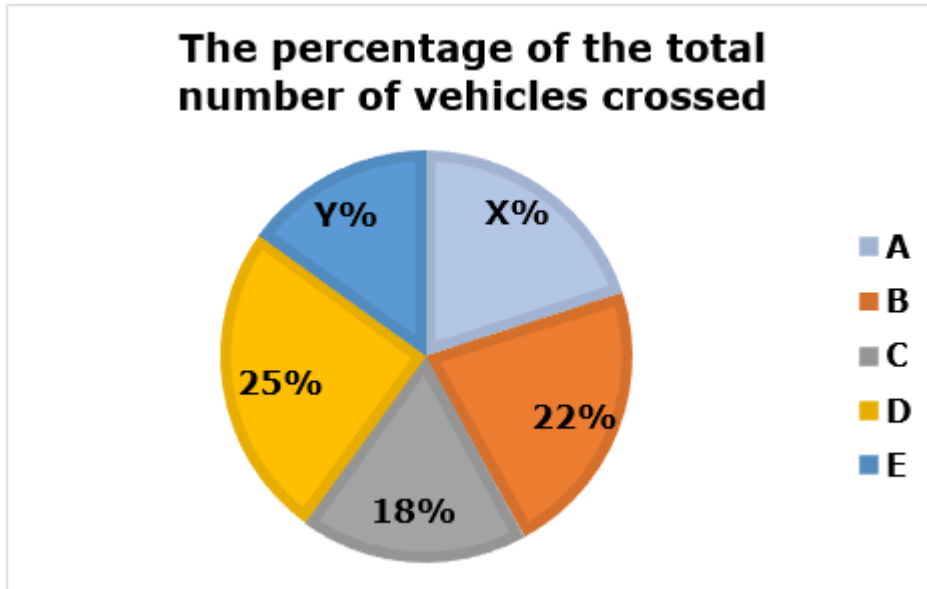
- a. 52.5%
- b. 59.75%
- c. 56.25%
- d. 47.25%

e. 43.75%

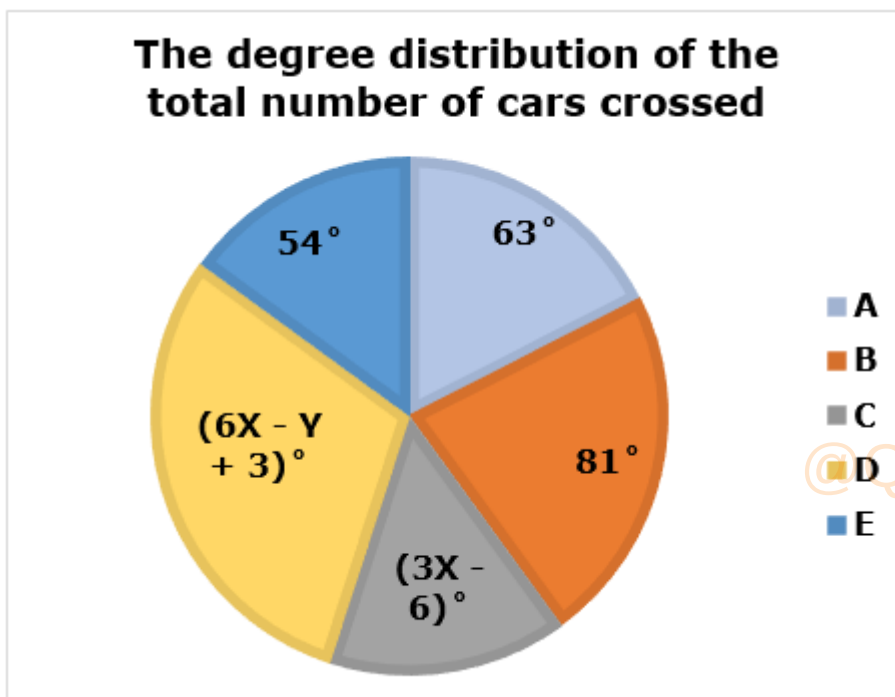
16. Questions

Study the following information carefully and answer the questions given below:

The below pie chart shows the percentage of the total number of vehicles crossed out of the total number of vehicles crossed in five toll plazas together from 9:00 AM to 10:00 AM.



The below pie chart shows the degree distribution of the total number of cars crossed out of the total number of cars crossed in five toll plazas together from 9:00 AM to 10:00 AM.



Notes:

i). The difference between the total number of vehicles crossed in toll plazas A and E from 9:00 AM to 10:00 AM is 40.

ii). The number of cars crossed in toll plaza D is $(3.6Y)\%$ of the total number of vehicles crossed in toll plaza D from 9:00 AM to 10:00 AM.

If the number of heavy vehicles crossed the toll plaza A is $X\%$ of the total vehicles crossed in the toll plaza and the remaining vehicles crossed in toll plaza A are bikes. Find the difference between the number of cars and that of bikes crossed in toll plaza A from 9:00 AM to 10:00 AM.

- a. 2
- b. 4
- c. 12
- d. 15
- e. 5

17. Questions

If the rate of a single journey and a monthly pass to a car in toll plaza B is Rs. 80 and Rs. 1800, respectively. The total revenue generated from the number of cars crossed in toll plaza B from 9:00 AM to 10:00 AM is Rs. 40880. Find the number of cars had a monthly pass that crossed the toll plaza B from 9:00 AM to 10:00 AM.

- a. 40
- b. 10
- c. 20
- d. 25
- e. 15

18. Questions

If the toll plaza C has 5 lanes and the ratio of the number of vehicles crossed in the first three lanes from 9:00 AM to 10:00 AM is 5: 4: 2, respectively, If the number of cars crossed only in the last two lanes and the number of vehicles other than cars in the first three lanes is 42 more than that of the last two lanes. Find the number of vehicles other than cars crossed in the last two lanes.

- a. $X + 4$
- b. $Y + 9$
- c. $X + Y - 11$
- d. All of them
- e. None of them

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19. Questions

If the number of cars crossed the toll plaza D from 10.00 AM to 11.00 AM and that of 11.00 AM to 12.00 PM is $5Y\%$ and 66.66% from the previous time slot, respectively. Find the average number

of cars that crossed the toll plaza from 9:00 AM to 12:00 PM.

- a. It is a perfect biquadrate
- b. It is a perfect cube
- c. It is a perfect square
- d. All of them
- e. Both (a) and (c)

20. Questions

The number of vehicles other than cars that crossed all the toll plazas together is what percentage of the total number of vehicles that crossed all the toll plazas together from 9:00 AM to 10:00 AM?

- a. $(2X + 5)\%$
- b. $(2X + Y)\%$
- c. $(2Y + 5)\%$
- d. $(2X + 2Y)\%$
- e. $(2Y + X)\%$

Explanations:

1. Questions

Answer: C

The ratio of the present age of A to that of B = 9: 14

The ratio of the present age of B to that of C = 4: 5

The ratio of the present age of A to that of B to that of C

36: 56: 70 = 18: 28: 35

Let the ages of A, B, and C be $18x$, $28x$ and $35x$

The possible difference between the present ages of A and C is 17 years and 34 years

(If the difference between the present ages of A and C is 17 years)

$$35x - 18x = 17$$

$$17x = 17$$

$$x = 1$$

So, the present age of A = $18x = 18$ years

The present age of B = $28x = 28$ years

Thus, average = $(18 + 28)/2 = 23$ years

If the difference between the present ages of A and C is 34 years

$$35x - 18x = 34$$

$$17x = 34$$

$$x = 2$$

So, the present age of A = $18x = 36$ years

The present age of B = $28x = 56$ years

Thus, average = $(36 + 56)/2 = 46$ years

Hence, Options I and III.

2. Questions

Answer: A

Arun lend amount to John = Rs. 21000

Let Interest rate = $R\%$ monthly

$N = 24$ months

$$SI = PNR/100$$

$$21000 * R * 24/100 = 28560 - 21000$$

$$210 * 24 * R = 7560$$

$$R = 1.5$$

The amount invested in scheme A = Rs. 28560

$$R = 10R = 15\%$$

$$N = 2$$

The interest received from scheme A

$$= 28560 * (115/100)^2 - 28560$$

$$= 28560 * (23^2 - 20^2)/20^2$$

$$= 71.4 * 129$$

$$= 9210.6$$

Thus, required difference = $9210.6 - 7560 = \text{Rs. } 1650.6$

3. Questions

Answer: B

Let the salaries of Raja and Ravi in 2018 be $5x$ and $4x$ respectively.

The salary of Raja in 2019 = $(5x/3) * 4 = 20x/3$

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The salary of Ravi in 2019 = $(4x/6) * 7 = 14x/3$

$$(1/5) * (20x/3) + (1/7) * (14x/3) = 6000$$

$$4x/3 + 2x/3 = 6000$$

$$6x/3 = 6000$$

$$x = 3000$$

The difference between the salaries of Raja and Ravi in 2018 = $5x - 4x = x = \text{Rs. } 3000$

Let

I). Multiple of 8

Let

$$3000/8 = 375$$

This is satisfied.

II). Multiple of 1000

Let

$$3000/1000 = 3$$

This is satisfied.

III). Multiple of 2000

Let

$$3000/2000 = 1.5$$

This is not satisfied.

IV). Multiple of 3

Let

$$3000/3 = 1000$$

This is satisfied.

4. Questions

Answer: D

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Factors of 8 are 1, 2, 4, and 8

A, B, C, and D to complete the work in 8 days, 4 days, 2 days, and 1 day

Total work = 8 units (LCM of 1, 2, 4 and 8)

The efficiency of A = $8/8 = 1$

The efficiency of B = $8/4 = 2$

The efficiency of C = $8/2 = 4$

The efficiency of D = $8/1 = 8$

D completed the work in 1 day with half of his efficiency = $(8/2) * 1 = 4$ units

The remaining work = $8 - 4 = 4$ units

The time taken by remaining work completed by A, B, and C

= $4/(1 + 2 + 25\% \text{ of } 4) = 4/(3 + 1) = 4/4 = 1$

So, the total time taken to complete the work = $1 + 1 = 2$ days.

5. Questions

Answer: A

The speed of man = $2.5 * 18/5 = 9$ km/hr

I). 1.35, 9, and 720

The time taken = $1.35/9 = 0.15$ hour = $0.15 * 60 = 9$ minutes

The new speed of man = $9 + 1 = 10$ km/hr

Time taken = $2/10 = 0.2$ hour = $0.2 * 360 = 720$ seconds

This is satisfied.

II). 2.25, 15, and 720

The time taken = $2.25/9 = 0.25$ hour = $0.25 * 60 = 15$ minutes

The new speed of man = $9 + 1 = 10$ km/hr

Time taken = $2/10 = 0.2$ hour = $0.2 * 360 = 720$ seconds

This is satisfied.

III). 1.125, 9, and 360

The time taken = $1.125/9 = 0.125$ hour = $0.125 * 60 = 7.5$ minutes

$7.5 \neq 9$

This is not satisfied.

6. Questions

In shop A

The total number of earbuds headphones available = $X + 40$

The number of over-ear headphones available = $60\% \text{ of } (X + 40) = 0.6X + 24$

The number of in-ear headphones available = $(0.6X + 24)/12 * 7 = 0.35X + 14$

In shop D

The total number of earbuds headphones available = Y

The number of over-ear headphones available = $49\% \text{ of } Y = 0.49Y$

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The number of in-ear headphones available = $(0.49Y)/7 * 5 = 0.35Y$

Given that, The number of in-ear headphones available in shop A is 21 less than that of shop D.

$$0.35Y - 0.35X - 14 = 21$$

$$0.35Y - 0.35X = 35$$

$$Y - X = 100 \text{ à (1)}$$

In shop C

The total number of earbuds headphones available = $X - 20$

The number of over-ear headphones available = 55% of $(X - 20) = 0.55X - 11$

The number of in-ear headphones available = $(0.55X - 11)/11 * 6 = 0.3X - 6$

The total number of headphones available = $X - 20 + 0.55X - 11 + 0.3X - 6 = 1.85X - 37$

The total number of headphones available in shop C is 333.

$$1.85X - 37 = 333$$

$$1.85X = 370$$

$$X = 200$$

From (1)

$$Y - X = 100$$

$$Y = 200 + 100$$

$$Y = 300$$

Let

In shop A

The total number of earbuds headphones available = $X + 40 = 240$

The number of over-ear headphones available = 60% of 240 = 144

The number of in-ear headphones available = $144/12 * 7 = 84$

The total number of headphones available = $240 + 144 + 84 = 468$

In shop B

The total number of earbuds headphones available = $Y - 45 = 255$

The number of over-ear headphones available = (40% of X)% of 255 = 80% of 255 = 204

The number of in-ear headphones available = $204/17 * 10 = 120$

The total number of headphones available = $255 + 204 + 120 = 579$

In shop C

The total number of earbuds headphones available = $X - 20 = 180$

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The number of over-ear headphones available = 55% of 180 = 99

The number of in-ear headphones available = $99/11 * 6 = 54$

The total number of headphones available = 333

In shop D

The total number of earbuds headphones available = $Y = 300$

The number of over-ear headphones available = 49% of 300 = 147

The number of in-ear headphones available = $147/7 * 5 = 105$

The total number of headphones available = $300 + 147 + 105 = 552$

Let table

Shop	The number of earbuds headphones available	The number of over-ear headphones available	The number of in-ear headphones available	The total number of headphones available
A	240	144	84	468
B	255	204	120	579
C	180	99	54	333
D	300	147	105	552

Answer: E

According to the question

The difference between the number of in-ear headphones available in shop A and the number of over-ear headphones available in shop B = $204 - 84 = 120$

$P = 120$

a). The number of earbuds headphones available in shop A is $2P$

$2P = 240$

$2 * 120 = 240$

This is satisfied.

b). The number of earbuds headphones available in shop C is $1.5P$

$1.5P = 180$

$1.5 * 120 = 180$

$180 = 180$

This is satisfied.

c). The number of in-ear headphones available in shop B is P

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$$120 = 120$$

This is satisfied.

Hence, option e

7. Questions

In shop A

The total number of earbuds headphones available = $X + 40$

The number of over-ear headphones available = 60% of $(X + 40) = 0.6X + 24$

The number of in-ear headphones available = $(0.6X + 24)/12 * 7 = 0.35X + 14$

In shop D

The total number of earbuds headphones available = Y

The number of over-ear headphones available = 49% of $Y = 0.49Y$

The number of in-ear headphones available = $(0.49Y)/7 * 5 = 0.35Y$

Given that, The number of in-ear headphones available in shop A is 21 less than that of shop D.

$$0.35Y - 0.35X - 14 = 21$$

$$0.35Y - 0.35X = 35$$

$$Y - X = 100 \text{ à (1)}$$

In shop C

The total number of earbuds headphones available = $X - 20$

The number of over-ear headphones available = 55% of $(X - 20) = 0.55X - 11$

The number of in-ear headphones available = $(0.55X - 11)/11 * 6 = 0.3X - 6$

The total number of headphones available = $X - 20 + 0.55X - 11 + 0.3X - 6 = 1.85X - 37$

The total number of headphones available in shop C is 333.

$$1.85X - 37 = 333$$

$$1.85X = 370$$

$$X = 200$$

From (1)

$$Y - X = 100$$

$$Y = 200 + 100$$

$$Y = 300$$

Let

In shop A

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The total number of earbuds headphones available = $X + 40 = 240$

The number of over-ear headphones available = 60% of $240 = 144$

The number of in-ear headphones available = $144/12 * 7 = 84$

The total number of headphones available = $240 + 144 + 84 = 468$

In shop B

The total number of earbuds headphones available = $Y - 45 = 255$

The number of over-ear headphones available = $(40\% \text{ of } X)\%$ of $255 = 80\%$ of $255 = 204$

The number of in-ear headphones available = $204/17 * 10 = 120$

The total number of headphones available = $255 + 204 + 120 = 579$

In shop C

The total number of earbuds headphones available = $X - 20 = 180$

The number of over-ear headphones available = 55% of $180 = 99$

The number of in-ear headphones available = $99/11 * 6 = 54$

The total number of headphones available = 333

In shop D

The total number of earbuds headphones available = $Y = 300$

The number of over-ear headphones available = 49% of $300 = 147$

The number of in-ear headphones available = $147/7 * 5 = 105$

The total number of headphones available = $300 + 147 + 105 = 552$

Let table

Shop	The number of earbuds headphones available	The number of over-ear headphones available	The number of in-ear headphones available	The total number of headphones available
A	240	144	84	468
B	255	204	120	579
C	180	99	54	333
D	300	147	105	552

Answer: A

The number of earbuds headphones unsold = 40% of $255 = 102$

The number of over-ear headphones unsold = 25% of $204 = 51$

The number of in-ear headphones unsold = 20% of 120 = 24

The total number of unsold headphones in shop B = 102 + 51 + 24 = 177

8. Questions

In shop A

The total number of earbuds headphones available = $X + 40$

The number of over-ear headphones available = 60% of $(X + 40) = 0.6X + 24$

The number of in-ear headphones available = $(0.6X + 24)/12 * 7 = 0.35X + 14$

In shop D

The total number of earbuds headphones available = Y

The number of over-ear headphones available = 49% of $Y = 0.49Y$

The number of in-ear headphones available = $(0.49Y)/7 * 5 = 0.35Y$

Given that, The number of in-ear headphones available in shop A is 21 less than that of shop D.

$$0.35Y - 0.35X - 14 = 21$$

$$0.35Y - 0.35X = 35$$

$$Y - X = 100 \text{ à (1)}$$

In shop C

The total number of earbuds headphones available = $X - 20$

The number of over-ear headphones available = 55% of $(X - 20) = 0.55X - 11$

The number of in-ear headphones available = $(0.55X - 11)/11 * 6 = 0.3X - 6$

The total number of headphones available = $X - 20 + 0.55X - 11 + 0.3X - 6 = 1.85X - 37$

The total number of headphones available in shop C is 333.

$$1.85X - 37 = 333$$

$$1.85X = 370$$

$$X = 200$$

From (1)

$$Y - X = 100$$

$$Y = 200 + 100$$

$$Y = 300$$

Let

In shop A

The total number of earbuds headphones available = $X + 40 = 240$

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The number of over-ear headphones available = 60% of 240 = 144

The number of in-ear headphones available = $144/12 * 7 = 84$

The total number of headphones available = $240 + 144 + 84 = 468$

In shop B

The total number of earbuds headphones available = $Y - 45 = 255$

The number of over-ear headphones available = (40% of X)% of 255 = 80% of 255 = 204

The number of in-ear headphones available = $204/17 * 10 = 120$

The total number of headphones available = $255 + 204 + 120 = 579$

In shop C

The total number of earbuds headphones available = $X - 20 = 180$

The number of over-ear headphones available = 55% of 180 = 99

The number of in-ear headphones available = $99/11 * 6 = 54$

The total number of headphones available = 333

In shop D

The total number of earbuds headphones available = $Y = 300$

The number of over-ear headphones available = 49% of 300 = 147

The number of in-ear headphones available = $147/7 * 5 = 105$

The total number of headphones available = $300 + 147 + 105 = 552$

Let table

Shop	The number of earbuds headphones available	The number of over-ear headphones available	The number of in-ear headphones available	The total number of headphones available
A	240	144	84	468
B	255	204	120	579
C	180	99	54	333
D	300	147	105	552

Answer: C

Quantity I:

The average of the number of in-ear headphones available in shops A, B, and C together = $(84 + 120 + 54)/3 = 258/3 = 86$

Quantity II:

The defective headphone in shop D = $(100 - 83.33)\%$ of 552 = $(1/6) * 552 = 92$

Thus, Quantity I < Quantity II

9. Questions

In shop A

The total number of earbuds headphones available = $X + 40$

The number of over-ear headphones available = 60% of $(X + 40) = 0.6X + 24$

The number of in-ear headphones available = $(0.6X + 24)/12 * 7 = 0.35X + 14$

In shop D

The total number of earbuds headphones available = Y

The number of over-ear headphones available = 49% of $Y = 0.49Y$

The number of in-ear headphones available = $(0.49Y)/7 * 5 = 0.35Y$

Given that, The number of in-ear headphones available in shop A is 21 less than that of shop D.

$$0.35Y - 0.35X - 14 = 21$$

$$0.35Y - 0.35X = 35$$

$$Y - X = 100 \text{ à (1)}$$

In shop C

The total number of earbuds headphones available = $X - 20$

The number of over-ear headphones available = 55% of $(X - 20) = 0.55X - 11$

The number of in-ear headphones available = $(0.55X - 11)/11 * 6 = 0.3X - 6$

The total number of headphones available = $X - 20 + 0.55X - 11 + 0.3X - 6 = 1.85X - 37$

The total number of headphones available in shop C is 333.

$$1.85X - 37 = 333$$

$$1.85X = 370$$

$$X = 200$$

From (1)

$$Y - X = 100$$

$$Y = 200 + 100$$

$$Y = 300$$

Let

In shop A

The total number of earbuds headphones available = $X + 40 = 240$

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The number of over-ear headphones available = 60% of 240 = 144

The number of in-ear headphones available = $144/12 * 7 = 84$

The total number of headphones available = $240 + 144 + 84 = 468$

In shop B

The total number of earbuds headphones available = $Y - 45 = 255$

The number of over-ear headphones available = (40% of X)% of 255 = 80% of 255 = 204

The number of in-ear headphones available = $204/17 * 10 = 120$

The total number of headphones available = $255 + 204 + 120 = 579$

In shop C

The total number of earbuds headphones available = $X - 20 = 180$

The number of over-ear headphones available = 55% of 180 = 99

The number of in-ear headphones available = $99/11 * 6 = 54$

The total number of headphones available = 333

In shop D

The total number of earbuds headphones available = $Y = 300$

The number of over-ear headphones available = 49% of 300 = 147

The number of in-ear headphones available = $147/7 * 5 = 105$

The total number of headphones available = $300 + 147 + 105 = 552$

Let table

Shop	The number of earbuds headphones available	The number of over-ear headphones available	The number of in-ear headphones available	The total number of headphones available
A	240	144	84	468
B	255	204	120	579
C	180	99	54	333
D	300	147	105	552

Answer: D

The number of earbuds headphones sold = $6x$

The number of over-ear headphones sold = x

The number of in-ear headphones sold = $119 - x$

Let

$$6x * 500 + x * 800 + (119 - x) * 850 = 204400$$

$$60x + 16x + 2023 - 17x = 4088$$

$$59x = 2065$$

$$x = 35$$

$$\text{The total number of headphones sold} = 6x + x + 119 - x = 210 + 119 = 329$$

$$\text{Thus, percentage} = (329/552) * 100 \approx 60\%$$

10. Questions

In shop A

$$\text{The total number of earbuds headphones available} = X + 40$$

$$\text{The number of over-ear headphones available} = 60\% \text{ of } (X + 40) = 0.6X + 24$$

$$\text{The number of in-ear headphones available} = (0.6X + 24)/12 * 7 = 0.35X + 14$$

In shop D

$$\text{The total number of earbuds headphones available} = Y$$

$$\text{The number of over-ear headphones available} = 49\% \text{ of } Y = 0.49Y$$

$$\text{The number of in-ear headphones available} = (0.49Y)/7 * 5 = 0.35Y$$

Given that, The number of in-ear headphones available in shop A is 21 less than that of shop D.

$$0.35Y - 0.35X - 14 = 21$$

$$0.35Y - 0.35X = 35$$

$$Y - X = 100 \text{ à (1)}$$

In shop C

$$\text{The total number of earbuds headphones available} = X - 20$$

$$\text{The number of over-ear headphones available} = 55\% \text{ of } (X - 20) = 0.55X - 11$$

$$\text{The number of in-ear headphones available} = (0.55X - 11)/11 * 6 = 0.3X - 6$$

$$\text{The total number of headphones available} = X - 20 + 0.55X - 11 + 0.3X - 6 = 1.85X - 37$$

The total number of headphones available in shop C is 333.

$$1.85X - 37 = 333$$

$$1.85X = 370$$

$$X = 200$$

From (1)

$$Y - X = 100$$

$$Y = 200 + 100$$

$$Y = 300$$

Let

In shop A

$$\text{The total number of earbuds headphones available} = X + 40 = 240$$

$$\text{The number of over-ear headphones available} = 60\% \text{ of } 240 = 144$$

$$\text{The number of in-ear headphones available} = 144/12 * 7 = 84$$

$$\text{The total number of headphones available} = 240 + 144 + 84 = 468$$

In shop B

$$\text{The total number of earbuds headphones available} = Y - 45 = 255$$

$$\text{The number of over-ear headphones available} = (40\% \text{ of } X)\% \text{ of } 255 = 80\% \text{ of } 255 = 204$$

$$\text{The number of in-ear headphones available} = 204/17 * 10 = 120$$

$$\text{The total number of headphones available} = 255 + 204 + 120 = 579$$

In shop C

$$\text{The total number of earbuds headphones available} = X - 20 = 180$$

$$\text{The number of over-ear headphones available} = 55\% \text{ of } 180 = 99$$

$$\text{The number of in-ear headphones available} = 99/11 * 6 = 54$$

$$\text{The total number of headphones available} = 333$$

In shop D

$$\text{The total number of earbuds headphones available} = Y = 300$$

$$\text{The number of over-ear headphones available} = 49\% \text{ of } 300 = 147$$

$$\text{The number of in-ear headphones available} = 147/7 * 5 = 105$$

$$\text{The total number of headphones available} = 300 + 147 + 105 = 552$$

Let table

@QuizByMickey

Shop	The number of earbuds headphones available	The number of over-ear headphones available	The number of in-ear headphones available	The total number of headphones available
A	240	144	84	468
B	255	204	120	579
C	180	99	54	333
D	300	147	105	552

Answer: E

The total number of headphones available in shop E = 133.33% of 333 = $(\frac{4}{3}) * 333 = 444$

The number of earbuds headphones available in shop E = $(0.5 * 200 - 20)\%$ of 240 = 80% of 240 = 192

The number of in-ear headphones available in shop E = 122.22% of 54 = $(\frac{11}{9}) * 54 = 66$

The number of over-ear headphones available in shop E = $444 - 192 - 66 = 186$

11. Questions

From line graph

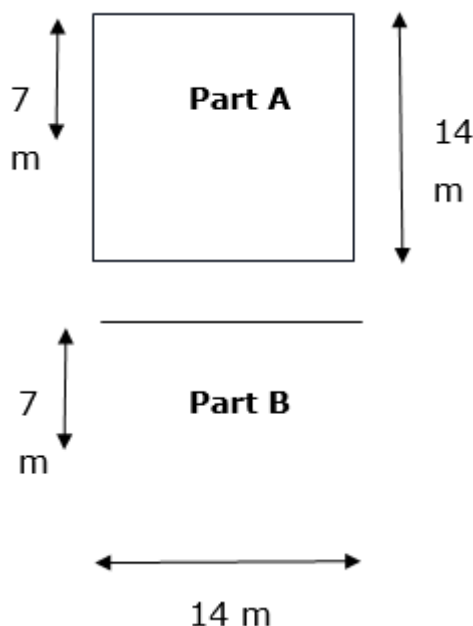
$\frac{1}{7}$ of the perimeter of the square plot = 8

$$(\frac{1}{7}) * 4a = 8$$

$$a = 14$$

The side of the square plot = 14 m

The side of the square plot is divided as two equal parts. So, two rectangles are formed, So, part A is rectangle shaped



@QuizByMickey

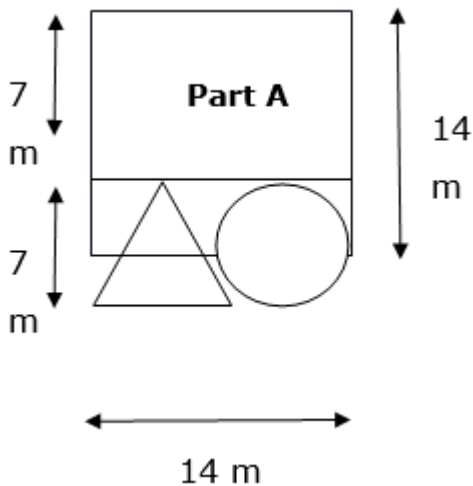
The base of the equatorial triangle-shaped garden is half of the side of the square plot.

So, the base of the equatorial triangle-shaped garden = $14/2 = 7$ m

The breadth of part B is the same as the diameter of the circle-shaped walking spot.

The breadth of the part B = 7 m

The diameter of the circle-shaped walking spot = 7 m



Q = The perimeter of the circle-shaped walking spot = $2 * (22/7) * 7/2 = 22$ m

Q = 22

P = the perimeter of the part A = $2 * (14 + 7) = 42$ m

P = 42

R = the perimeter of the equatorial triangle-shaped garden = $3 * 7 = 21$ m

R = 21

Then

X = P - R

X = 42 - 21

X = 21

Y = Q - R

Y = 22 - 21

Y = 1

Z = P - Q

Z = 42 - 22

Z = 20

Answer: A

According to the question

@QuizByMickey

The total area of part B = $7 * 14 = 98 \text{ m}^2$

The area of the circle-shaped walking spot = $(22/7) * 3.5 * 3.5 = 38.5 \text{ m}^2$

The area of the equatorial triangle-shaped garden = $(\sqrt{3}/4) * 7^2 = 12.25 * 1.7 = 20.825 \text{ m}^2$

The remaining area of part B = $98 - 38.5 - 20.825 = 38.675 \text{ m}^2$

12. Questions

From line graph

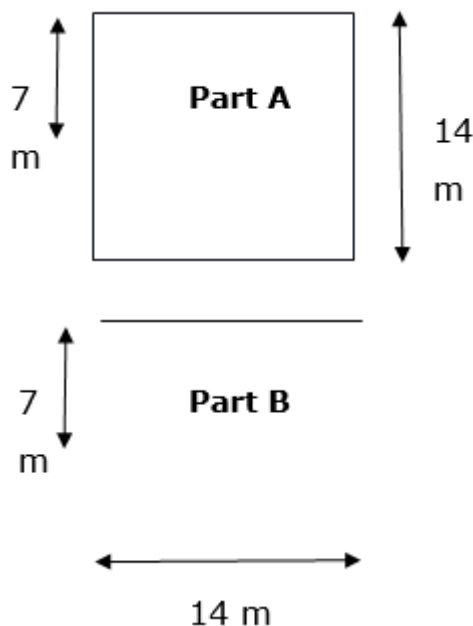
$1/7$ of the perimeter of the square plot = 8

$(1/7) * 4a = 8$

$a = 14$

The side of the square plot = 14 m

The side of the square plot is divided as two equal parts. So, two rectangles are formed, So, part A is rectangle shaped



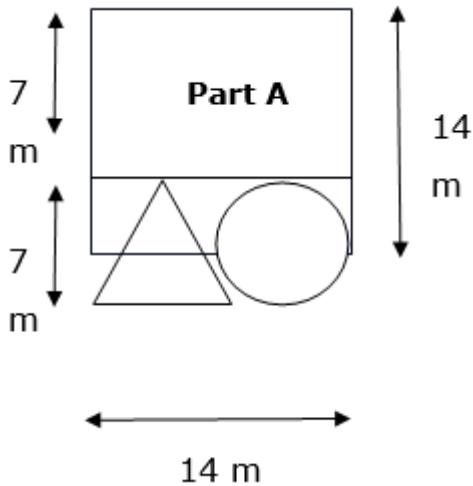
The base of the equatorial triangle-shaped garden is half of the side of the square plot.

So, the base of the equatorial triangle-shaped garden = $14/2 = 7 \text{ m}$

The breadth of part B is the same as the diameter of the circle-shaped walking spot.

The breadth of the part B = 7 m

The diameter of the circle-shaped walking spot = 7 m



Q = The perimeter of the circle-shaped walking spot = $2 * (22/7) * 7/2 = 22$ m

Q = 22

P = the perimeter of the part A = $2 * (14 + 7) = 42$ m

P = 42

R = the perimeter of the equatorial triangle-shaped garden = $3 * 7 = 21$ m

R = 21

Then

$X = P - R$

$X = 42 - 21$

$X = 21$

$Y = Q - R$

$Y = 22 - 21$

$Y = 1$

$Z = P - Q$

$Z = 42 - 22$

$Z = 20$

Answer: E

The radius of the water tank = 150% of 14 = 21 m

The volume of the cylinder-shaped water tank = $\pi r^2 h$

$(22/7) * 21^2 * h = 33264$

$3 * 21 * h = 1512$

$h = 24$

@QuizByMickey

The height of the water tank is 24 m

13. Questions

From line graph

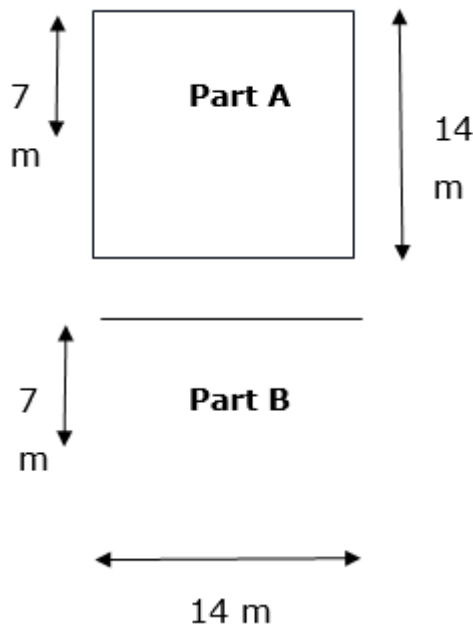
$\frac{1}{7}$ of the perimeter of the square plot = 8

$$\left(\frac{1}{7}\right) * 4a = 8$$

$$a = 14$$

The side of the square plot = 14 m

The side of the square plot is divided as two equal parts. So, two rectangles are formed, So, part A is rectangle shaped



The base of the equatorial triangle-shaped garden is half of the side of the square plot.

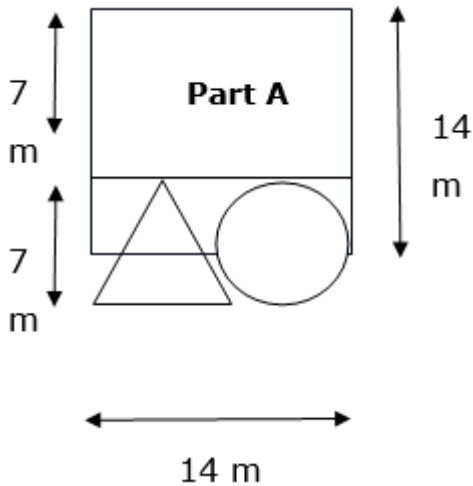
So, the base of the equatorial triangle-shaped garden = $\frac{14}{2} = 7$ m

The breadth of part B is the same as the diameter of the circle-shaped walking spot.

The breadth of the part B = 7 m

The diameter of the circle-shaped walking spot = 7 m

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$Q = \text{The perimeter of the circle-shaped walking spot} = 2 * (22/7) * 7/2 = 22 \text{ m}$

$Q = 22$

$P = \text{the perimeter of the part A} = 2 * (14 + 7) = 42 \text{ m}$

$P = 42$

$R = \text{the perimeter of the equatorial triangle-shaped garden} = 3 * 7 = 21 \text{ m}$

$R = 21$

Then

$X = P - R$

$X = 42 - 21$

$X = 21$

$Y = Q - R$

$Y = 22 - 21$

$Y = 1$

$Z = P - Q$

$Z = 42 - 22$

$Z = 20$

Answer: D

Let

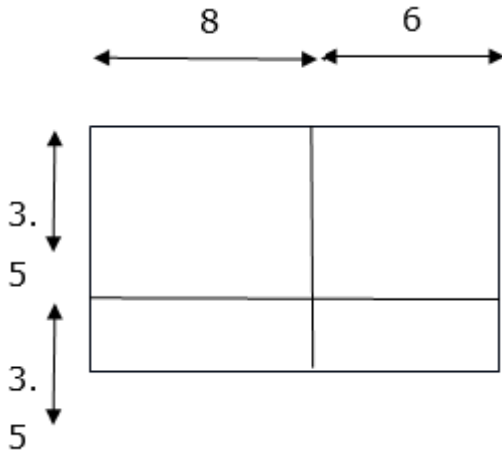
The breadth of part A $= 7/2 = 3.5 \text{ m}$

The total length of part A $= 14 \text{ m}$

The length of the larger housing plot in part A $= (14/7) * 4 = 8 \text{ m}$

Let diagram

@QuizByMickey



Then

The area of the larger housing plot in part A = $8 * 3.5 = 28 \text{ m}^2$

14. Questions

From line graph

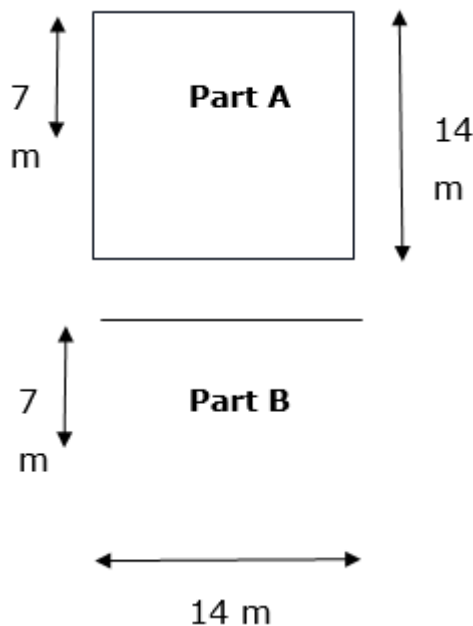
$\frac{1}{7}$ of the perimeter of the square plot = 8

$$\left(\frac{1}{7}\right) * 4a = 8$$

$$a = 14$$

The side of the square plot = 14 m

The side of the square plot is divided as two equal parts. So, two rectangles are formed, So, part A is rectangle shaped



@QuizByMickey

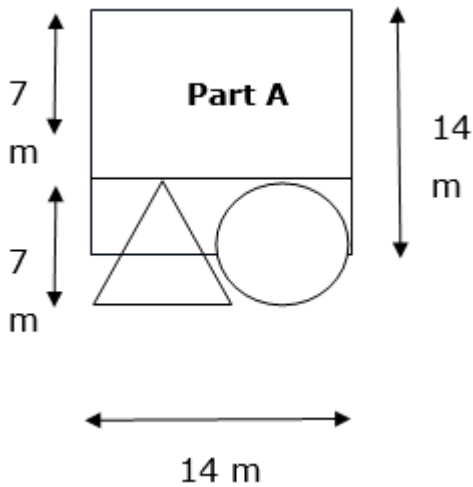
The base of the equatorial triangle-shaped garden is half of the side of the square plot.

So, the base of the equatorial triangle-shaped garden = $\frac{14}{2} = 7 \text{ m}$

The breadth of part B is the same as the diameter of the circle-shaped walking spot.

The breadth of the part B = 7 m

The diameter of the circle-shaped walking spot = 7 m



Q = The perimeter of the circle-shaped walking spot = $2 * (22/7) * 7/2 = 22$ m

Q = 22

P = the perimeter of the part A = $2 * (14 + 7) = 42$ m

P = 42

R = the perimeter of the equatorial triangle-shaped garden = $3 * 7 = 21$ m

R = 21

Then

$X = P - R$

$X = 42 - 21$

$X = 21$

$Y = Q - R$

$Y = 22 - 21$

$Y = 1$

$Z = P - Q$

$Z = 42 - 22$

$Z = 20$

Answer: B

The area of part A = $7 * 14 = 98 \text{ m}^2$

The area of the circle-shaped walking spot in part B = $(22/7) * 3.5^2 = 38.5 \text{ m}^2$

Thus, required ratio = $98: 38.5 = 28: 11$

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15. Questions

From line graph

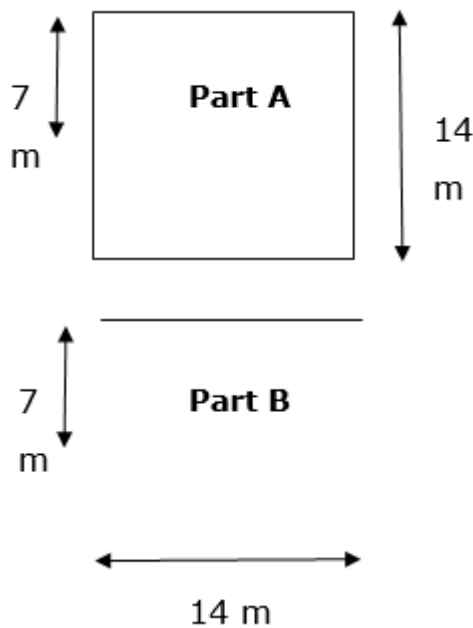
$\frac{1}{7}$ of the perimeter of the square plot = 8

$$\left(\frac{1}{7}\right) * 4a = 8$$

$$a = 14$$

The side of the square plot = 14 m

The side of the square plot is divided as two equal parts. So, two rectangles are formed, So, part A is rectangle shaped



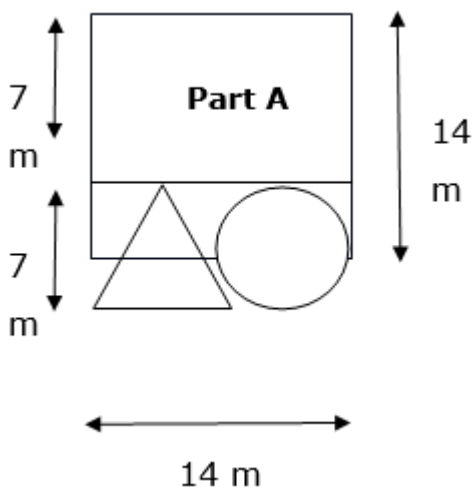
The base of the equatorial triangle-shaped garden is half of the side of the square plot.

So, the base of the equatorial triangle-shaped garden = $\frac{14}{2} = 7$ m

The breadth of part B is the same as the diameter of the circle-shaped walking spot.

The breadth of the part B = 7 m

The diameter of the circle-shaped walking spot = 7 m



@QuizByMickey

Q = The perimeter of the circle-shaped walking spot = $2 * (22/7) * 7/2 = 22$ m

$$Q = 22$$

P = the perimeter of the part A = $2 * (14 + 7) = 42$ m

$$P = 42$$

R = the perimeter of the equatorial triangle-shaped garden = $3 * 7 = 21$ m

$$R = 21$$

Then

$$X = P - R$$

$$X = 42 - 21$$

$$X = 21$$

$$Y = Q - R$$

$$Y = 22 - 21$$

$$Y = 1$$

$$Z = P - Q$$

$$Z = 42 - 22$$

$$Z = 20$$

Answer: C

The new side of the square plot = 125% of 14 = 17.5 m

The new area of the square plot = $17.5 * 17.5 = 306.25 \text{ m}^2$

Thus, required percentage = $[306.25 - 196]/196 * 100 = 56.25\%$

16. Questions

From pie chart 1

$$X + 22 + 18 + 25 + Y = 100$$

$$X + Y = 100 - 65$$

$$X + Y = 35 \text{ ---> (1)}$$

From pie chart 2

$$63 + 81 + 3X - 6 + 6X - Y + 3 + 54 = 360$$

$$9X - Y = 360 - 195$$

$$9X - Y = 165 \text{ ---> (2)}$$

Solve (1) and (2)

$$X + 9X = 35 + 165$$

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$$10X = 200$$

$$X = 20$$

$$20 + Y = 35$$

$$Y = 15$$

The total number of vehicles crossed in five toll plazas together = $100a$

The total number of vehicles crossed in toll plaza A = $X\%$ of $100a = 20\%$ of $100a = 20a$

The total number of vehicles crossed in toll plaza E = $Y\%$ of $100a = 15\%$ of $100a = 15a$

Then

$$20a - 15a = 40$$

$$5a = 40$$

$$a = 8$$

The total number of vehicles crossed in five toll plazas together = $100a = 100 * 8 = 800$

The total number of vehicles crossed in toll plaza A = 20% of $800 = 160$

The total number of vehicles crossed in toll plaza B = 22% of $800 = 176$

The total number of vehicles crossed in toll plaza C = 18% of $800 = 144$

The total number of vehicles crossed in toll plaza D = 25% of $800 = 200$

The total number of vehicles crossed in toll plaza E = 15% of $800 = 120$

Then

The number of cars crossed in toll plaza D is $(3.6Y)\%$ of the total number of vehicles crossed in toll plaza D.

The number of cars crossed in toll plaza D = $3.6 * 15\%$ of $200 = 54\%$ of $200 = 108$

The degree distribution of the number of cars crossed in toll plaza D = $6X - Y + 3 = 120 - 15 + 3 = 108$

The number of cars crossed in toll plaza A = $(108/108^\circ) * 63^\circ = 63$

The number of cars crossed in toll plaza B = $(108/108^\circ) * 81^\circ = 81$

The number of cars crossed in toll plaza C = $(108/108^\circ) * (3X - 6)^\circ = (108/108^\circ) * 54^\circ = 54$

The number of cars crossed in toll plaza E = $(108/108^\circ) * 54^\circ = 54$

Let table

Toll plaza	The total number of vehicles crossed	The number of cars crossed
A	160	63
B	176	81
C	144	54
D	200	108
E	120	54

Answer: A

According to the question

The number of heavy vehicles crossed the toll plaza A = $X\%$ of 160 = 20% of 160 = 32

The number of cars crossed the toll plaza A = 63

The number of bikes crossed the toll plaza A = $160 - 32 - 63 = 65$

Thus, required difference = $65 - 63 = 2$

17. Questions

From pie chart 1

$$X + 22 + 18 + 25 + Y = 100$$

$$X + Y = 100 - 65$$

$$X + Y = 35 \rightarrow (1)$$

From pie chart 2

$$63 + 81 + 3X - 6 + 6X - Y + 3 + 54 = 360$$

$$9X - Y = 360 - 195$$

$$9X - Y = 165 \rightarrow (2)$$

Solve (1) and (2)

$$X + 9X = 35 + 165$$

$$10X = 200$$

$$X = 20$$

$$20 + Y = 35$$

$$Y = 15$$

The total number of vehicles crossed in five toll plazas together = $100a$

The total number of vehicles crossed in toll plaza A = $X\%$ of $100a = 20\%$ of $100a = 20a$

The total number of vehicles crossed in toll plaza E = $Y\%$ of $100a = 15\%$ of $100a = 15a$

Then

@QuizByMickey

$$20a - 15a = 40$$

$$5a = 40$$

$$a = 8$$

The total number of vehicles crossed in five toll plazas together = $100a = 100 * 8 = 800$

The total number of vehicles crossed in toll plaza A = 20% of 800 = 160

The total number of vehicles crossed in toll plaza B = 22% of 800 = 176

The total number of vehicles crossed in toll plaza C = 18% of 800 = 144

The total number of vehicles crossed in toll plaza D = 25% of 800 = 200

The total number of vehicles crossed in toll plaza E = 15% of 800 = 120

Then

The number of cars crossed in toll plaza D is $(3.6Y)\%$ of the total number of vehicles crossed in toll plaza D.

The number of cars crossed in toll plaza D = $3.6 * 15\%$ of 200 = 54% of 200 = 108

The degree distribution of the number of cars crossed in toll plaza D = $6X - Y + 3 = 120 - 15 + 3 = 108$

The number of cars crossed in toll plaza A = $(108/108^\circ) * 63^\circ = 63$

The number of cars crossed in toll plaza B = $(108/108^\circ) * 81^\circ = 81$

The number of cars crossed in toll plaza C = $(108/108^\circ) * (3X - 6)^\circ = (108/108^\circ) * 54^\circ = 54$

The number of cars crossed in toll plaza E = $(108/108^\circ) * 54^\circ = 54$

Let table

Toll plaza	The total number of vehicles crossed	The number of cars crossed
A	160	63
B	176	81
C	144	54
D	200	108
E	120	54

Answer: C

The number of cars (monthly pass) that crossed the toll plaza B = b

the number of cars (single journey) that crossed the toll plaza B = $81 - b$

Let

$$(81 - b) * 80 + b * 1800 = 40880$$

$$6480 - 80b + 1800b = 40880$$

$$1720b = 34400$$

$$b = 20$$

18. Questions

From pie chart 1

$$X + 22 + 18 + 25 + Y = 100$$

$$X + Y = 100 - 65$$

$$X + Y = 35 \rightarrow (1)$$

From pie chart 2

$$63 + 81 + 3X - 6 + 6X - Y + 3 + 54 = 360$$

$$9X - Y = 360 - 195$$

$$9X - Y = 165 \rightarrow (2)$$

Solve (1) and (2)

$$X + 9X = 35 + 165$$

$$10X = 200$$

$$X = 20$$

$$20 + Y = 35$$

$$Y = 15$$

The total number of vehicles crossed in five toll plazas together = $100a$

The total number of vehicles crossed in toll plaza A = $X\%$ of $100a = 20\%$ of $100a = 20a$

The total number of vehicles crossed in toll plaza E = $Y\%$ of $100a = 15\%$ of $100a = 15a$

Then

$$20a - 15a = 40$$

$$5a = 40$$

$$a = 8$$

The total number of vehicles crossed in five toll plazas together = $100a = 100 * 8 = 800$

The total number of vehicles crossed in toll plaza A = 20% of $800 = 160$

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The total number of vehicles crossed in toll plaza C = 18% of $800 = 144$

The total number of vehicles crossed in toll plaza D = 25% of $800 = 200$

The total number of vehicles crossed in toll plaza E = 15% of $800 = 120$

Then

@QuizByMickey

The number of cars crossed in toll plaza D is $(3.6Y)\%$ of the total number of vehicles crossed in toll plaza D.

The number of cars crossed in toll plaza D = $3.6 \times 15\%$ of 200 = 54% of 200 = 108

The degree distribution of the number of cars crossed in toll plaza D = $6X - Y + 3 = 120 - 15 + 3 = 108$

The number of cars crossed in toll plaza A = $(108/108^\circ) \times 63^\circ = 63$

The number of cars crossed in toll plaza B = $(108/108^\circ) \times 81^\circ = 81$

The number of cars crossed in toll plaza C = $(108/108^\circ) \times (3X - 6)^\circ = (108/108^\circ) \times 54^\circ = 54$

The number of cars crossed in toll plaza E = $(108/108^\circ) \times 54^\circ = 54$

Let table

Toll plaza	The total number of vehicles crossed	The number of cars crossed
A	160	63
B	176	81
C	144	54
D	200	108
E	120	54

Answer: D

The number of cars crossed only in the last two lanes only

The number of cars crossed the toll plaza C = 54

So, the number of vehicles other than cars crossed the toll plaza C = $144 - 54 = 90$

Then

The number of vehicles crossed in the first three lanes = The number of vehicles other than cars crossed in the first three lanes

The number of vehicles other than cars crossed in the first three lanes + the number of vehicles other than cars crossed in the last two lanes = 90

The number of vehicles other than cars crossed in the first three lanes - the number of vehicles other than cars crossed in the last two lanes = 42

The number of vehicles other than cars crossed in the first three lanes = $(90 + 42)/2 = 132/2 = 66$

The number of vehicles other than cars crossed in the last two lanes = $(90 - 42)/2 = 48/2 = 24$

a). $X + 4$

Let

$X + 4 = 24$

$24 = 24$

This is satisfied.

b). $Y + 9$

Let

$$Y + 9 = 24$$

$$15 + 9 = 24$$

$$24 = 24$$

This is satisfied.

c). $X + Y - 11$

Let

$$20 + 15 - 11 = 24$$

$$24 = 24$$

This is satisfied.

19. Questions

From pie chart 1

$$X + 22 + 18 + 25 + Y = 100$$

$$X + Y = 100 - 65$$

$$X + Y = 35 \rightarrow (1)$$

From pie chart 2

$$63 + 81 + 3X - 6 + 6X - Y + 3 + 54 = 360$$

$$9X - Y = 360 - 195$$

$$9X - Y = 165 \rightarrow (2)$$

Solve (1) and (2)

$$X + 9X = 35 + 165$$

$$10X = 200$$

$$X = 20$$

$$20 + Y = 35$$

$$Y = 15$$

The total number of vehicles crossed in five toll plazas together = $100a$

The total number of vehicles crossed in toll plaza A = $X\%$ of $100a = 20\%$ of $100a = 20a$

The total number of vehicles crossed in toll plaza E = $Y\%$ of $100a = 15\%$ of $100a = 15a$

Then

@QuizByMickey

$$20a - 15a = 40$$

$$5a = 40$$

$$a = 8$$

The total number of vehicles crossed in five toll plazas together = $100a = 100 * 8 = 800$

The total number of vehicles crossed in toll plaza A = 20% of 800 = 160

The total number of vehicles crossed in toll plaza B = 22% of 800 = 176

The total number of vehicles crossed in toll plaza C = 18% of 800 = 144

The total number of vehicles crossed in toll plaza D = 25% of 800 = 200

The total number of vehicles crossed in toll plaza E = 15% of 800 = 120

Then

The number of cars crossed in toll plaza D is $(3.6Y)\%$ of the total number of vehicles crossed in toll plaza D.

The number of cars crossed in toll plaza D = $3.6 * 15\%$ of 200 = 54% of 200 = 108

The degree distribution of the number of cars crossed in toll plaza D = $6X - Y + 3 = 120 - 15 + 3 = 108$

The number of cars crossed in toll plaza A = $(108/108^\circ) * 63^\circ = 63$

The number of cars crossed in toll plaza B = $(108/108^\circ) * 81^\circ = 81$

The number of cars crossed in toll plaza C = $(108/108^\circ) * (3X - 6)^\circ = (108/108^\circ) * 54^\circ = 54$

The number of cars crossed in toll plaza E = $(108/108^\circ) * 54^\circ = 54$

Let table

Toll plaza	The total number of vehicles crossed	The number of cars crossed
A	160	63
B	176	81
C	144	54
D	200	108
E	120	54

Answer: E

The number of cars crossed the toll plaza D from 10.00 AM to 11.00 AM = $5Y\%$ of 108 = 75% of 108 = 81

The number of cars crossed the toll plaza D from 11.00 AM to 12.00 PM = 66.66% of 81 = $(2/3) * 81 = 54$

Thus, average = $(108 + 81 + 54)/3 = 243/3 = 81$

$3^4 = 81$ (It is a perfect biquadrate)

$9^2 = 81$ (It is a perfect square)

20. Questions

From pie chart 1

$$X + 22 + 18 + 25 + Y = 100$$

$$X + Y = 100 - 65$$

$$X + Y = 35 \rightarrow (1)$$

From pie chart 2

$$63 + 81 + 3X - 6 + 6X - Y + 3 + 54 = 360$$

$$9X - Y = 360 - 195$$

$$9X - Y = 165 \rightarrow (2)$$

Solve (1) and (2)

$$X + 9X = 35 + 165$$

$$10X = 200$$

$$X = 20$$

$$20 + Y = 35$$

$$Y = 15$$

The total number of vehicles crossed in five toll plazas together = 100a

The total number of vehicles crossed in toll plaza A = X% of 100a = 20% of 100a = 20a

The total number of vehicles crossed in toll plaza E = Y% of 100a = 15% of 100a = 15a

Then

$$20a - 15a = 40$$

$$5a = 40$$

$$a = 8$$

The total number of vehicles crossed in five toll plazas together = 100a = 100 * 8 = 800

The total number of vehicles crossed in toll plaza A = 20% of 800 = 160

The total number of vehicles crossed in toll plaza B = 22% of 800 = 176

The total number of vehicles crossed in toll plaza C = 18% of 800 = 144

The total number of vehicles crossed in toll plaza D = 25% of 800 = 200

The total number of vehicles crossed in toll plaza E = 15% of 800 = 120

Then

The number of cars crossed in toll plaza D is (3.6Y)% of the total number of vehicles crossed in toll plaza D.

The number of cars crossed in toll plaza D = $3.6 \times 15\%$ of 200 = 54% of 200 = 108

The degree distribution of the number of cars crossed in toll plaza D = $6X - Y + 3 = 120 - 15 + 3 = 108$

The number of cars crossed in toll plaza A = $(108/108^\circ) \times 63^\circ = 63$

The number of cars crossed in toll plaza B = $(108/108^\circ) \times 81^\circ = 81$

The number of cars crossed in toll plaza C = $(108/108^\circ) \times (3X - 6)^\circ = (108/108^\circ) \times 54^\circ = 54$

The number of cars crossed in toll plaza E = $(108/108^\circ) \times 54^\circ = 54$

Let table

Toll plaza	The total number of vehicles crossed	The number of cars crossed
A	160	63
B	176	81
C	144	54
D	200	108
E	120	54

Answer: B

The total number of vehicles that crossed all the toll plazas together = $160/20 \times 100 = 800$

The number of cars that crossed all the toll plazas together = $81/81^\circ \times 360^\circ = 360$

The number of vehicles other than cars that crossed all the toll plazas together = $800 - 360 = 440$

Thus, percentage = $(440/800) \times 100 = 55\%$

$2X + Y = 40 + 15 = 55$

@QuizByMickey